

Generation Cardinality as a Forced Role-Occupancy Endpoint

Primitive necessity, exact-three occupation, and nonpromotion under enrichment

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Primitive-standing reference edition

Abstract

We prove that complete primitive generation standing is available exactly at three occurrence positions. The original task is specified with multiplicity open: representation response, charge and weak-chiral compatibility, mass-response slots, direct flavor comparison, boundary reconstruction, nonvacuous holonomy, selector independence, and joint realization. Primitive standing means that all this work is jointly constructible and no proper occurrence support suffices for the same work. The definition contains no preferred count or assumed circuit rank.

Complete comparison modulo vertex rephasing has non-removable circuit capacity precisely when its carrier contains a triple. Every minimal live support is a triangle. A uniform construction realizes every finite phase class in one common gauge, charged-response, and coherent-boundary packet, supplying all original duties jointly on every such support. Hence every primitive architecture has exactly three positions, including exclusion of all larger finite and infinite architectures from primitive standing.

The role template is formed from that derived architecture before candidate matching. Complete occupation of the full live union closes exactly at three, and its circuit register has rank one as a consequence. Downstream enlargement cannot acquire additional primitive generation authority for the unchanged task. A receiving relation gives the same primitive count without selecting a preferred embedded triangle; preservation of other original answers has its own exact fibre criterion. Grounded derived constructions preserve the original qualification closure.

The result fixes primitive necessity prior to downstream values and dynamics. Additional physical response content retains its declared standing without changing the arity of the complete primitive construction. The paper supplies the original-use ledger, positive joint receipt, no-surplus occupation proof, nonpromotion theorem, and receiving conditions for downstream applications.

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1 Primitive generation standing

The generation problem asks which occurrence architectures have *primitive standing*: they jointly perform the complete original generation work, and no proper occurrence support can perform that same work. The target is fixed before observed families, parameter values, or candidate multiplicities are supplied. The result is

$$\text{complete primitive generation standing at cardinality } n \iff n = 3.$$

The proof excludes every other primitive cardinality. It also establishes that downstream enlargement supplies no additional primitive role for the unchanged generation task.

1.1 Original work and inherited licensing

The original work comprises representation-response eligibility, charge and weak-chiral compatibility, mass-response slots, direct flavor comparison, boundary reconstruction, nonvacuous holonomy, selector independence, and joint realization. Definition 2.1 states these duties with multiplicity open. Section 2.1 specifies their original questions, domains, and receiving expressions.

The inherited matter, charge, weak and bridge types are received from the carrier sources [16, 14, 15]. The present domain construction receives those licenses and proves simultaneous compatibility, primitive adequacy, and the resulting occurrence count. Displayed matrices witness the compatibility equations; their entries do not independently create the inherited licenses.

A primitive duty is necessary licensing work. Numerical masses, matrix coordinates, calibrated phases, and particular propagation responses can be standing-bearing content of a realized object without themselves selecting a new primitive role. This distinction is part of the original generation problem [12, Definitions 4.3–5.1 and Section 1.3] and its carrier imports [16, Definitions 6.5–6.7].

1.2 The dependency direction

Non-degenerate determinate objects, incidences, and endpoints necessarily incur the Kernel’s work of admissibility, standing, reference, and irreversibility [4]. The generation theorem instantiates that necessary work through its comparison structure, joint construction, and complete occupation. Its cardinality is a domain result derived from those duties.

The primitive target precedes its enriched receivers. Write n for the number of occurrence positions in one primitive architecture and N for the ambient positions of a receiving object. An enlargement does not acquire additional primitive standing merely by supporting more interactions or responses. Its primitive count is the count of the architecture it receives. A claim identifying all its ambient positions with primitive occupants requires a complete occupation map.

The AMetric, complete-interface, exhaustion, and closure results retain the answers and grounds of an independently fixed target [5, 6, 7, 9]. The original target here is the primitive generation work. Further questions of an enriched physical task retain their own meanings and standing. The count is established by the domain construction; generic uniqueness of an interface or interior is not used as its numerical premise [8].

1.3 Proof map and public results

1. Fix the complete original duty and question inventory with candidate multiplicity open.

2. Prove that complete comparison modulo vertex rephasing first has nonvacuous circuit capacity on three positions. Every minimal live support is a triangle.
3. Construct one common receiving packet for each finite admitted phase class. This realizes all the duties together and supplies a joint receipt on every minimal live support.
4. Classify primitive necessity before naming roles: every minimal adequate architecture has three positions. Then form its role template before matching candidates and prove exact complete occupation.
5. Prove that enlarged realizations cannot add primitive positions to the unchanged task. Establish count descent through functional or relational receipts and conservation of original authority under grounded derived construction.

Theorem 5.4 gives exact-three closed occupation. Theorem 6.2 states the architectural necessity result directly, without putting three into the definition of primitive standing. Theorem 6.4 excludes promotion of enlarged architectures, and Theorems 6.6 and 6.8 state the corresponding receiving and authority results. The primitive circuit rank is one as a consequence of the derived architecture.

1.4 Constructional necessity and fixed-object identity

The constructional necessity test asks whether the same complete primitive work admits a joint realization on a proper support. Receiving parameters may be supplied together on that support. A transformation of one already fixed physical packet has a different obligation: it must preserve the original answers of that packet. The latter can include numerical values, full response histories, and additional continuations.

The count proof uses joint construction before those additional answers are fixed. It does not extract a dynamically autonomous triangle from every enriched receiver. Nor does it erase a response needed by the original primitive task. Original questions and their interpretation are fixed by the duty ledger before the minimum is calculated.

Several supports may be alternative complete primitive witnesses. A lawful architecture isomorphism retains their occurrence incidence. It does not identify different physical phase values or select one preferred embedded witness. Whole-template uniqueness and uniqueness of a realized parameter point are distinct statements.

1.5 Physical receiving family

The exhibited family uses the count-open matter representation types, a common Higgs bridge, charged Yukawa response, and an admitted law-only neutrino boundary option. Section 4 provides the actual algebraic and coherent-boundary equations. The inherited representation inventory is an input to this domain construction; numerical parameters establish existence, without observed masses or family labels selecting the count.

The architectural theorem concerns admitted capability. A live witness establishes that capability, while an actual CP-conserving parameter point may occupy the same architecture. The two points and their respective outcomes are kept separate in Section 6.

2 The primitive duty package before counting

The inherited generation carrier types are held fixed while their multiplicity remains open. The duty package $\mathcal{P}_0$ below expands the original symbols $R, Q, W, M, F, B, \Omega, S, J$ into work to be realized jointly. A phase comparison class is part of a constructional witness; empirical phase values are not inputs selecting the role architecture.

Definition 2.1 (The joint primitive generation duty package). A construction of $\mathcal{P}_0$ has anonymous occurrence incidence and the following capabilities, with an actual joint witness.

R, Q, W

It carries the inherited matter representation, charge, and weak-chiral types on anonymous multiplicity spaces. The gauge and bridge contractions are well typed, and the stated local and global charge/anomaly conditions hold together.

M The inherited bridge licenses charged response slots sufficient to support the claimed comparison incidences. Specific mass eigenvalues and their ordering are not fixed original answers of this primitive task.

F Distinct proposed generation occurrences participate in direct co-comparison. The comparison graph is the complete simple graph of those occurrences. Its phase descriptions are identified under vertex-local rephasing; no path, basis entry, or fitted zero replaces the direct comparison incidence.

B Source preparation, propagation, and endpoint readout have determinate compatible incidence in one declared boundary model. Flavor labels are not silently transported as conserved interior identities where the response does not commute with the endpoint projectors.

Ω The comparison architecture admits a non-removable closed-circuit consequence. The admitted phase family used here also supplies a reversal-sensitive witness; this is a capability assertion, not the assertion that every non-removable phase value is reversal sensitive. A capacity claim requires an admitted witnessing use; it does not require nonzero CP at every parameter point.

S The architecture, role formation, quotient, and candidate tests are defined before matching observed families. They do not receive an empirical count, preferred circuit, mass ordering, or fitted texture as selection authority.

J The foregoing work is jointly realizable with one common parameter packet and consistent source, boundary, response, and failure data. Its anonymous role assignments are interpreted covariantly. Separate successes in mutually incompatible packets do not establish joint adequacy.

Section 4 realizes Definition 2.1 through explicit carriers, maps, equations, and a common witness. Its direct-comparison quotient is received exactly, including the phase data and occurrence incidence. The physical receiver has further well-defined outputs; those outputs are retained as its own physical data without being used to select the primitive role count.

Duty	Source meaning received	What must be checked in the witness
R, Q, W	Original role certificate; inherited matter/charge/weak ledger	Representation modules, charge contractions, common multiplicity and anomaly equations
M, F	Mass-response-slot and transition licensing	Positive admissible responses, intrinsic occurrences, direct comparison and redescription covariance
B	Boundary reconstruction, without hidden flavor-name transport	One source/propagation/readout construction with compatible endpoint operators
Ω	Holonomy surviving phase-coordinate redescription	Exact phase quotient and a non-removable circuit witness
S	No empirical, dimension-display, or representative selector	Candidate-independent duties, all minimal supports, and lawful role transport
J	Whole-assignment compatibility and coverage	One common packet; actual eligibility, live occupants, and assignment action

The source anchors are the original Definitions 4.3–5.1, Sections 6–7, and Theorem 11.1 [12]; the flavor carrier definitions and boundary discipline [15, Sections 5–6 and 11]; and target-before-matching, whole-assignment compatibility, and completeness in the role-occupancy framework [10, Definitions 3.5–3.18 and 7.1]. SM-1 distinguishes licensing work from derivative values [16, Definitions 6.5–6.7]. These imports specify the work whose occurrence cardinality is derived below.

Direct comparison is a substantive hypothesis. The complete simple comparison carrier is the generation-incidence normal form used here. Its completeness is essential: on a sparse graph, a simple cycle of any length can be a minimal nonzero circuit. Physical phase channels also require realization. The uniform receiver supplies a corresponding intrinsic response for every declared direct edge and phase in this normal form. A downstream domain with a different comparison graph requires its own normal-form and minimality proofs.

Standing and numerical values. A numerical coordinate can carry standing-bearing physical information within a realized packet. An interface preserving that packet must retain its original answers [13]. The primitive minimum problem instead fixes the duties in Definition 2.1 and asks which occurrence architecture jointly realizes them. Additional numerical data remain part of the richer task in which they are specified.

Capacity and a witnessed use. The formal capacity predicate is introduced in Section 3. For the phase circle, ordinary nonzero capacity and reversal-sensitive capacity have the same least architecture. An exhibited live use establishes the capacity’s nonvacuity. A later CP-conserving value need not change the architecture or the number of primitive roles; it changes the status of that particular CP-sensitive use. The structural theorem therefore concerns architectural capacity independently of an empirical CP measurement.

2.1 Original-use inventory and joint coverage

The nine duty names acquire mathematical content through a received type inventory and the following original questions. The inventory fixes the gauge representations, charge and chiral types, admitted bridge contractions, complete direct-comparison meaning, and coherent boundary model. It does not fix occurrence multiplicity, a rank-one register, a phase value, or a three-dimensional mixing matrix.

The required questions concern whether these capabilities and compatibilities are jointly realized. Numerical quantities used to witness them remain actual quantities of the receiver. Their values are not prescribed across all constructions of the primitive task.

Duty	Original question	Receiving data and verification
R, Q, W	Do the inherited types and contractions coexist with the required charge and anomaly conditions?	The typed representation modules, Yukawa contractions, common multiplicity equations, local anomaly sums, and declared weak/global checks in Section 4
M	Are there compatible nondegenerate charged response slots for the claimed comparisons?	Positive simple responses A, B, C , their square-root Yukawa maps, and the one inherited bridge receipt
F	Does every declared direct pair have the certified comparison, with the actual phase quotient retained?	$\text{Tr}(P_i B P_j B)$ and the intrinsic loop readers of Proposition 4.3, including their incidence
B	Do the preparation, propagation and endpoint questions share one boundary model with actual domains?	Charged-lepton boundary projections E_i , one $U(t)$, and $\text{Tr}(E_j U(t) E_i U(t)^*)$ on the declared parameter domain
Ω	Is a non-removable circuit capability actually witnessed?	An admitted class q_* , its nonzero triangle consequence, and the test $2q_* \neq 0$ when reversal sensitivity is claimed
S	Do role formation and tests precede empirical matching and respect lawful relabeling?	Count-open definitions, the complete witness family, quotient transport, and absence of a selected empirical or embedded-triangle input
J	Are all the answers and demanded successful uses jointly realized?	One common packet for the complete assignment, typed joint domains, actual failure outcomes, and the generated-use proof of Proposition 4.5

The P_i are intrinsic quark occurrence projections of A ; the E_i are charged-lepton boundary projections on the lepton copy. Shared display indices do not supply an extra physical pairing map between quark and lepton eigenlines. Boundary transfer is required on the admitted witnessing domain established in Section 4; nonzero transition probability at every time, including zero time, is not an original obligation.

The inherited licensing premise and the constructive receipt play different parts. This construction receives the type and role licenses specified in SM-1/SM-3/SM-5. Section 4 proves that the displayed packet satisfies their stated joint compatibility equations at open multiplicity. The law-only neutral option is received from PCR-5 [11, Section 18]. No displayed matrix entry is treated as an independent proof of upstream source formation.

Finite derived uses. Take the actual packet maps, pair and loop observations, and boundary maps in the table as a many-sorted inventory. Each has an input type, output type, admission domain and a recorded failure outcome. Required successful calls retain their readiness conditions. Close this inventory under finite typed composition, contraction, adjoint, endomorphism trace, and inverse on its actual domain. The original derived questions of this construction are specified by expressions in that grammar and their independently stated meanings.

For each table row, the receiving expression or predicate supplies the original answer: representation and compatibility equations for R, Q, W, M ; a reader for each direct pair and each loop consequence for F, Ω ; source-propagation-readout for B ; and the common-incidence, transport and whole-packet conditions for S, J . Proposition 4.5 then proves domain and outcome preservation for every finite derived expression by induction on its construction. This gives a use-level receipt of the specified original task, not merely a list of duty names.

This coverage does not impose all numerical outputs of every enriched receiver as fixed answers of the primitive minimum problem. A new acceptance equation, infinitary obligation, or eventual-success condition requires its own original-task status and proof. Merely using the same underlying matrices does not make a new requirement a consequence of the prior task. The coverage criterion is the original-question discipline of fixed-domain exhaustion [7]; its generation-specific witnesses are the constructions above and in Section 4.

3 Primitive minimality before downstream enrichment

Minimality is measured against the independently fixed primitive work. The construction first determines its adequate architectures; downstream enrichment supplies further numerical outputs, dynamics, and continuations on a realized architecture.

3.1 Capacity, live realizations, and minimal architecture

For an occurrence set I and abelian group B , put

$$C_I^0(B) = B^I, \quad (d\phi)_{ij} = \phi_j - \phi_i, \\ Q_I(B) = C_I^1(B)/dC_I^0(B), \quad T(q)_{ijk} = c_{ij} + c_{jk} + c_{ki}.$$

Cochains are skew, vanish on the diagonal, and may have arbitrary support. The phase group is written additively as $\mathbb{R}/\mathbb{Z}$ or multiplicatively as $U(1)$ through $t \mapsto e^{2\pi it}$; reversal is negation or inversion, respectively. The triangle tensor detects the quotient: $T(q) = 0$ if and only if $q = 0$. For nonempty I , this follows by fixing a temporary vertex o and using $\phi_i = c_{oi}$ when all triangle sums vanish. The choice is confined to the proof.

Definition 3.1 (Ordinary and reversal-sensitive primitive capacity). Define

$$\text{Cap}_B(I) \iff Q_I(B) \neq 0, \quad \text{Cap}_B^-(I) \iff \exists q \in Q_I(B), 2q \neq 0.$$

A capacity-bearing architecture is inclusion-minimal if no proper subset retains the stated capacity. These conditions mention neither a preferred count nor an empirical phase value.

Theorem 3.2 (Minimal complete-comparison capacity). *For every occurrence set I :*

$$\text{Cap}_B(I) \iff (B \neq 0 \text{ and } I \text{ contains a triple}),$$

and

$$\text{Cap}_B^-(I) \iff (2B \neq 0 \text{ and } I \text{ contains a triple}).$$

If $B \neq 0$, every inclusion-minimal ordinary-capacity architecture has exactly three positions. If $2B \neq 0$, the same is true for reversal-sensitive capacity. The least possible cardinality in each nonempty class is three, and its bare complete graph has one isomorphism type, K_3 .

Proof. On at most two positions every cochain is a vertex gradient. If $B = 0$, the quotient is zero. If $2B = 0$, every quotient class is killed by two. Conversely, choose distinct a, b, c and a coefficient $b_0 \neq 0$, or $2b_0 \neq 0$ respectively. Put b_0 on the oriented edge ab and zero on all other edges. Its triangle response on abc is b_0 , so the required capacity exists.

Any capacity-bearing I contains a capacity-bearing triple. Thus a larger I is not inclusion-minimal; a triple is minimal because all its proper subsets have zero capacity. This argument also excludes infinite minimal architectures. $\square$

Capacity belongs to the declared comparison architecture and coefficient type. A later phase choice $q = 0$ or $2q = 0$ retains that capacity while giving a CP-conserving realized point. Standing of a particular reversal-sensitive use depends on the realized class.

Theorem 3.3 (Pointwise live-support minimality). *Let $q \in Q_I(B)$ with $2q \neq 0$. The following are equivalent:*

1. *no proper subset $J \subsetneq I$ has $2q|_J \neq 0$;*
2. *for every $v \in I$, $2q|_{I \setminus \{v\}} = 0$;*
3. *$|I| = 3$.*

Proof. A triangle detects the nonzero class $2q$, so some triple J has $2q|_J \neq 0$. If I has an occurrence v outside J , deleting v leaves that live triangle. Thus (2) implies (3), and (1) implies (2). If $|I| = 3$, every proper subset has zero comparison quotient, giving (1). $\square$

The quantifiers in indispensability. Each occurrence participating in *some* indispensable local witness does not imply Theorem 3.3(2). Indeed, whenever $2q \neq 0$, every occurrence belongs to a live triangle. Given a live abc and an outside v , the identity

$$T(q)_{abc} = T(q)_{vbc} - T(q)_{vac} + T(q)_{vab}$$

shows that at least one triangle on the right is live. Every position of that triangle is indispensable to that local witness, even when the whole carrier has more than three positions.

Nor does the family-wide condition

$$\forall v \in I \exists q \in Q_I(B) : \quad 2q \neq 0, \quad 2q|_{I \setminus \{v\}} = 0$$

force three. For $|I| \geq 3$ and $2B \neq 0$, an edge cochain incident to v satisfies it. The occurrence is needed for *some* permitted input, although a particular larger construction can contain another live witness after its deletion.

3.2 Joint realization of the fixed primitive duties

Fix a primitive target $\mathcal{W}$, independently of candidate cardinality. Let $A_{\mathcal{W}}(I, q)$ mean that there exists a joint realization of its primitive duties on (I, q) . Additional realization data may be quantified existentially here. Adequacy on a smaller support asks for a realization of the same primitive target, with those additional data synthesized on that support.

Definition 3.4 (Primitive necessity and witness reception). The following are separate, inspectable domain assertions:

1. *Nonvacuity:* $A_{\mathcal{W}}(I, q) \Rightarrow 2q \neq 0$.
2. *Joint minimal-witness reception:* if $A_{\mathcal{W}}(I, q)$, then some inclusion-minimal live support $J \subseteq I$ admits $A_{\mathcal{W}}(J, q|_J)$.
3. *Inhabitation:* at least one joint realization exists.

In (2), all original primitive eligibility and joint-compatibility requirements must be satisfied. Duties belonging only to a downstream enrichment are not included. No selected witness or full enriched-output preservation map is asserted.

Theorem 3.5 (Minimal joint primitive construction). *Assume Definition 3.4. A joint realization $A_{\mathcal{W}}(I, q)$ is inclusion-minimal among its adequate primitive subrealizations if and only if $|I| = 3$. The class of joint realizations has least cardinality three. Every inclusion-minimal bare complete-comparison architecture is isomorphic to K_3 .*

Proof. Joint reception supplies a minimal live support J on which all the primitive duties are realizable. Theorem 3.3 gives $|J| = 3$. Inclusion-minimality of the joint realization therefore forces $I = J$. Conversely, every proper subset of a three-position realization has zero quotient, so nonvacuity excludes an adequate realization there. Inhabitation and reception produce a three-position realization; nonvacuity excludes fewer positions. The complete bare graph on three positions has one isomorphism type. $\square$

Joint reception preserves the *meaning and complete obligations* of $\mathcal{W}$. Holonomy nonvacuity supplies the live support; adequacy requires all primitive duties to be realized there together. Metric values, a full response algebra, and tasks added by an enrichment remain data of that enriched realization.

For a general joint predicate, testing only single-vertex deletion is insufficient unless an appropriate upward-closure property is proved. A smaller adequate support may exist even if intermediate supports fail a compatibility condition. Theorem 3.5 uses absence of *every* proper adequate primitive subrealization.

Proposition 3.6 (Minimum realization cardinality). *Let $A_{\mathcal{W}}$ be any class of realizations of one independently fixed primitive target. If it contains a three-position realization and none with fewer than three positions, then its minimum cardinality is three. This conclusion requires no restriction, compression, or reconstruction map from larger realizations. It does not alone imply that every realization is inclusion-minimal.*

Proof. The positive example gives an upper bound of three on the minimum; the exclusions give the matching lower bound. $\square$

Theorem 3.7 (Joint capacity before a realized phase choice). *Let $P_{\mathcal{W}}(I)$ mean that the bare occurrence architecture admits some joint realization of the independently specified primitive work. Assume:*

1. $P_{\mathcal{W}}$ is invariant under occurrence bijections;
2. $P_{\mathcal{W}}(I)$ implies $\text{Cap}_B^-(I)$;
3. one three-position architecture has a proved joint realization.

Then the inclusion-minimal $P_{\mathcal{W}}$ architectures are exactly the three-position architectures, all of one bare isomorphism type. Their minimum cardinality is three. This conclusion is independent of the subsequently realized phase point on such an architecture.

Proof. Nonvacuous capacity excludes fewer than three positions. The positive realization transports along every bijection to every three-position architecture. A larger adequate architecture contains a proper three-position adequate architecture, so is not inclusion-minimal. Three-position architectures have no smaller adequate support. $\square$

Here adequacy on the smaller architecture means existence of a joint realization of the same primitive work. The existence statement is independent of an extraction map from a fixed larger packet and of the CP value measured in a particular realization.

If the original role template is formed from these minimal joint capacity architectures, then the role arity is derived as three. For an input occurrence carrier I , all three-position embeddings admitted by that bare capability target have live union I when $|I| \geq 3$; when $|I| < 3$ there is no preassignment. Complete occupation by one template therefore occurs exactly at three. This capacity-level application needs the original certificate to concern primitive capability, rather than the success of a particular nonzero-phase trial. Those are distinct standing questions.

Theorem 3.8 (Joint reception and the closed endpoint). *Suppose the following claims have been proved about the original closed generation certificates:*

1. *their primitive joint predicate satisfies Definition 3.4(1)–(2);*
2. *no-surplus primitive work entails that their counted occurrence carrier is inclusion-minimal for that predicate;*
3. *a positive three-position original certificate satisfies all original eligibility, joint, and lawful occupancy conditions.*

Then

$$\text{ClosedGenCert}(n) \iff n = 3.$$

Alternatively, (1)–(2) may be replaced by an independently justified identification of the certified count with the least realization cardinality of the fixed primitive target, together with exclusion of all counts below three.

Proof. In the first route, any closed certificate is a minimal joint primitive realization, so Theorem 3.5 gives three. The positive certificate gives the converse. In the minimum-cardinality route use Proposition 3.6. $\square$

The count follows from three domain results: the independently established primitive duties, their joint reception, and the constructional meaning of no surplus. The Kernel supplies the necessary roles; the joint-realization and occupancy constructions below establish these occurrence-level premises.

Primitive minimality yields K_3 , whose cochain rank is one. Rank one is therefore a consequence of the primitive architecture theorem.

3.3 Joint primitive templates and complete occupancy

Theorem 3.9 (Uniform joint reception derives the role template). *Assume $2B \neq 0$ and an independently fixed primitive target $\mathcal{W}$. Suppose joint adequacy implies reversal-sensitive liveness and that every minimal live support of every live comparison class admits a joint realization of $\mathcal{W}$ with that restricted comparison class.*

Form the template class of inclusion-minimal joint adequate primitive completions, retaining their bare homogeneous occurrence positions. Then this class is nonempty and has exactly one bare architecture, with three positions. Let $R_{\mathcal{W}}$ represent this architecture, formed before any candidate residue matching.

For a live candidate $q \in Q_I(B)$, let all injections $\sigma : R_{\mathcal{W}} \rightarrow I$ whose images are joint adequate minimal primitive supports be the preassignments. Their live occupant union is I . Hence an injection from $R_{\mathcal{W}}$ covering the full live union exists if and only if $|I| = 3$. At three, if compatibility and the lawful action retain the stated bare homogeneous typing, all such assignments form one orbit under permutations of $R_{\mathcal{W}}$.

Proof. Theorem 3.5 shows that every minimal joint completion has three positions. Nonemptiness follows by putting a coefficient not killed by two on one triangle edge and using the joint receipt. Thus the template's arity is derived from primitive adequacy, not specified by a proposed count.

For a candidate q , nonvacuity and uniform reception make its adequate minimal supports exactly its live triangles. Their union is I by the tetrahedral coverage argument. Every such support is the image of a bijection from $R_{\mathcal{W}}$. A complete assignment is therefore an injection from a derived three-position set onto I , and exists exactly when $|I| = 3$. At three, precomposition by role permutations acts transitively on bijections while fixing all occupant data. $\square$

The template is the architecture of minimal adequate completions of the *complete primitive work*. Uniform joint reception determines its arity before candidate matching. If a target's role typing carries extra decorations, joint reception must also establish their eligibility, joint predicate, and lawful action. Bare architecture and decorated role typing are distinct levels of this construction.

The type of a constructional preassignment. The fixed problem in Theorem 3.9 has candidate (I, q) and the primitive duty specification $\mathcal{W}$. If nonphase realization data are existentially quantified, its joint predicate has the form

$$J_{\text{prim}}(\sigma) \iff \exists \eta \text{ Realizes}_{\mathcal{W}}(\text{im } \sigma, q|_{\text{im } \sigma}, \eta).$$

One witness η supplies one common packet for the whole assignment. Different preassignments can have different receipts within the same declared constructional problem. Their union consists of *constructionally eligible incidence positions*. At complete occupation with $|I| = 3$, one receipt covers the whole primitive carrier, and its invariant occupant construction supplies physical occupation within that packet.

For a physical parent x_N with fixed spectra, couplings, boundary records, or continuations, eligibility of its actual occurrences is a separate predicate. Establishing it requires a realization that respects those fixed data; uniform resynthesis establishes constructional eligibility. Receipts for alternative constructions are consequently not required to agree on unassigned overlaps.

An enriched ambient carrier may receive primitive witnesses through a realization relation. The primitive whole-union occupation condition applies to the minimal architecture, while additional ambient positions belong to the specified enrichment.

3.4 Architecture, selectors, and conservative enrichment

Bare minimal architecture uniqueness does not mean uniqueness of a realized holonomy value or of an embedded witness. For example, triangle values 1 and 2 over $\mathbb{Z}$ have the same bare architecture but are not related by occurrence permutation.

Proposition 3.10 (Equivariant selection obstruction). *There is no occurrence-equivariant rule selecting one embedded live triangle from every live integral cochain carrier. Nevertheless every such carrier has a minimal live witness and all minimal witness architectures have three positions.*

Proof. On four occurrences put coefficient 1 along the directed cycle $1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$, coefficient 0 on the two diagonals, and use skew symmetry. All four ascending triangle sums are 2. The cyclic permutation (1234) fixes this cochain class and acts without a fixed three-element subset. An equivariant selector would have to choose a subset fixed by that stabilizer, which is impossible. Existence and common witness arity follow from triangle detection. $\square$

It is therefore sufficient to retain the natural witness relation

$$\{(q, J) : J \text{ is an adequate minimal primitive witness for } q\} \longrightarrow \{q\},$$

with nonempty fibres when reception holds. Passing to the bare architecture gives one isomorphism type without supplying a preferred physical subcarrier. An existential proof uses a temporary witness; it does not license that witness as a selected original output.

Proposition 3.11 (Conservative enrichment is relative to primitive uses). *Let $\mathcal{E}$ be enriched objects, let $\mathcal{P}$ be primitive objects, and let $u : \mathcal{E} \twoheadrightarrow \mathcal{P}$ be a proposed surjective interpretation (replace $\mathcal{P}$ by the attained image when necessary). Fix the original primitive readers $h : \mathcal{E} \rightarrow Y_h$, including primitive admission domains. The target readers to be induced on $\mathcal{P}$ are not independently prescribed. The interpretation conserves those uses exactly when every such reader factors through u . No factorization of additional enriched readers follows or is needed. If primitive count is defined on $\mathcal{P}$, its value on an enriched realization is the count of $u(e)$, not automatically the cardinality of the enriched ambient carrier.*

Proof. Factorization says precisely that the original primitive answer is determined by the interpreted primitive object. Failure to factor gives two objects with the same interpretation and different original answers, witnessing a lost primitive distinction. Readers outside the fixed primitive family impose no equation in this assertion. The count statement follows by composing the primitive count with u . $\square$

The factorization criterion applies to each *proposed* interpretation and its declared primitive readers. A witness relation instead specifies which enriched receivers realize a primitive construction; functional factorization requires a separately established map. This distinction accommodates the selector obstruction while retaining the primitive and enriched objects' respective output identities.

4 Joint realization of the primitive generation duties

The construction realizes the received primitive matter, charge, weak, mass-response, flavor, and boundary duties in one shared packet. Representation and carrier licensing are the SM-1/SM-3/SM-5 inputs [16, 14, 15]. The law-only Weinberg option uses the admitted construction type of PCR-5 [11, Section 18]. Standard Yukawa and coherent-neutrino notation follows the corresponding reviews [1, 2, 3]. The equations are posed at arbitrary finite nonempty cardinality; numerical choices then witness joint compatibility. The structural role construction determines the arity independently of observed family counts and empirical parameter values.

4.1 A count-open shared packet

Proposition 4.1 (Shared multiplicity from the received charge inventory). *Allow independent nonnegative multiplicities n_Q, n_u, n_d, n_L, n_e of the received Standard Model representations and hypercharges. If their local gauge and mixed gravitational anomalies cancel, then*

$$n_Q = n_u = n_d = n_L = n_e.$$

This is a count-independent inventory constraint. It does not select the common value or pair quark and lepton mass eigenlines. An additional hypercharge-zero singlet multiplicity is not constrained by this argument.

Proof. Writing right-handed fields as left-handed conjugates, the $SU(3)^3$, $SU(3)^2Y$, $SU(2)^2Y$, and mixed gravitational conditions, after multiplication by their nonzero normalization constants, are

$$\begin{aligned} 2n_Q - n_u - n_d &= 0, \\ n_Q - 2n_u + n_d &= 0, \\ n_Q - n_L &= 0, \\ n_Q - 2n_u + n_d - n_L + n_e &= 0. \end{aligned}$$

The first two give $3n_Q - 3n_u = 0$, hence $n_u = n_Q$ and $n_d = n_Q$. The third gives $n_L = n_Q$, and the fourth then gives $n_e = n_Q$. The remaining cubic hypercharge condition and the even weak-doublet condition hold at every common multiplicity, as checked below. $\square$

Let I be any finite nonempty occurrence set, with $|I| = n$. Construct the standard left-doublet and right-singlet multiplicity spaces as copies of $\mathbb{C}^I$. Their gauge modules and hypercharges are

$$(\mathbb{C}^3 \otimes \mathbb{C}^2) \otimes V_Q (1/6), \quad \mathbb{C}^3 \otimes V_u (2/3), \quad \mathbb{C}^3 \otimes V_d (-1/3), \quad \mathbb{C}^2 \otimes V_L (-1/2), \quad V_e (-1).$$

The weak components of each doublet share one multiplicity space. Take one Higgs doublet of hypercharge $1/2$ and a common nonzero fixation $v > 0$. For example, a classical potential $\lambda(H^*H - v^2/2)^2$, $\lambda > 0$, supplies the common vacuum orbit. Gauge couplings and all remaining declared scalar parameters are fixed once for the whole packet. No additional neutral-fermion node is introduced.

Choose positive distinct a_i, c_i and real d_i with

$$d_i > n - 1, \quad |d_i - d_j| > 2(n - 1) \quad (i \neq j).$$

For a representative z of any $q \in Q_I(U(1))$, define

$$\begin{aligned} A &= \text{diag}(a_i), \quad C = \text{diag}(c_i), \quad B_{ii} = d_i, \quad B_{ij} = z_{ij}, \quad z_{ji} = \overline{z_{ij}}, \\ Y_u &= A^{1/2}, \quad Y_d = B^{1/2}, \quad Y_e = C^{1/2}. \end{aligned}$$

These are maps between the appropriate right and left multiplicity spaces. The displayed identifications of their copies are presentation data. Independent sector changes of basis transport every tensor and incidence by its actual left/right action.

Theorem 4.2 (Joint count-open capability realization). *For every finite nonempty I and every $q \in Q_I(U(1))$, the construction above has positive simple charged mass responses, all direct quark comparison responses nonzero, and a shared lawful gauge/charge/weak realization. There is a simultaneous law-only neutral packet with nontrivial boundary transfer on every pair of active channels when $n \geq 2$. The quark Dirac pair violates weak CP exactly when $2q \neq 0$. Thus every live phase class has a joint realization of the received primitive duties; no particular phase or mass values select the count.*

Proof. Positivity of B follows from

$$x^* B x \geq \sum_i (d_i - (n - 1)) |x_i|^2 > 0 \quad (x \neq 0).$$

Its disjoint Gershgorin intervals contain exactly one eigenvalue each, by continuous deformation from $\text{diag}(d_i)$. Thus B is simple and all charged maps are invertible. The choices $a_i = i^2$, $c_i = (2i - 1)^2$, $d_i = 3ni$ in any displayed enumeration show existence at every n ; the enumeration is not a retained ordering. If P_i are the spectral line projections of A , then

$$\text{Tr}(P_i B P_j B) = |B_{ij}|^2 = 1 \quad (i \neq j).$$

The certified co-comparability graph is therefore complete. Moreover A, B generate the full matrix algebra: spectral interpolation supplies P_i , and $P_i B P_j = B_{ij} E_{ij}$ supplies every matrix unit.

The three charged Yukawa contractions have hypercharge sums

$$-\frac{1}{6} - \frac{1}{2} + \frac{2}{3} = 0, \quad -\frac{1}{6} + \frac{1}{2} - \frac{1}{3} = 0, \quad \frac{1}{2} + \frac{1}{2} - 1 = 0.$$

With right-handed fields written as left-handed conjugates, the familywise local anomaly sums are

$$\begin{aligned} SU(3)^3 : 2 - 1 - 1 &= 0, \\ SU(3)^2 Y : 2(1/6)(1/2) - (2/3)(1/2) + (1/3)(1/2) &= 0, \\ SU(2)^2 Y : 3(1/6)(1/2) - (1/2)(1/2) &= 0, \\ Y^3 : 6(1/6)^3 - 3(2/3)^3 + 3(1/3)^3 - 2(1/2)^3 + 1 &= 0, \\ \text{grav}^2 Y : 1 - 2 + 1 - 1 + 1 &= 0. \end{aligned}$$

There are $4n$ left weak doublets. The usual central $\mathbb{Z}_6$ element acts trivially with integral charges $6Y$. These checks hold for every input cardinality and hence do not supply a numerical selector.

For the neutral packet take

$$N = \text{diag}(1, \dots, n) + \mathbf{1}\mathbf{1}^T, \quad m_\nu = \mu\sqrt{N}, \quad \kappa = 2m_\nu/v^2, \quad \mu > 0.$$

This is a real symmetric Weinberg coefficient on the same L, H fields. The positive matrix N has simple spectrum: the secular function $1 + \sum_{i=1}^n (i - \lambda)^{-1}$ is strictly increasing between its poles, with one zero in each $(i, i+1)$ and one above n ; no pole itself is an eigenvalue. In the declared coherent vacuum propagation model, put $t = \mu^2 L / (2E)$, $E > 0$, and $U(t) = e^{-itN}$. Let E_i be the charged-lepton source/detection projections on the lepton multiplicity space. They form a complete boundary interface, with $\text{Tr}(E_j U(t) E_i U(t)^*) = |U_{ji}(t)|^2$. These E_i are distinct typed objects from the quark occurrence projections P_i . Since $\|N\| \leq 2n$, the unitary integral remainder gives

$$\|U(t) - I + itN\| \leq 2n^2 t^2, \quad |U_{ji}(t)| \geq t - 2n^2 t^2 \geq t/2 > 0$$

for $i \neq j$ and $0 < t \leq 1/(4n^2)$. This supplies one common unitary kernel, rather than independent channel fits. The same kernel, its composition, its domains, and its endpoint projectors are retained in the packet. Flavor at a boundary is not asserted to be a conserved interior label.

Finally A is real and simple. A unitary weak-CP intertwiner must commute with A and hence is diagonal in its eigenframe. CP invariance of the pair is therefore equivalent to vertex-gauge equivalence of B and $\bar{B}$, that is, $q = -q$. Conversely, if $2q = 0$, gauge star edges to 1; every remaining edge is a real triangle holonomy, ± 1 . Both positive square-root Yukawas then have real representatives. Thus weak CP is violated exactly when $2q \neq 0$. $\square$

Proposition 4.3 (Intrinsic and faithful comparison quotient). *For the fixed nondegenerate A and declared diagonal data d_i , retain its occurrence projections P_i and their incidences. The full phase family above realizes exactly $Q_I(U(1))$ modulo its vertex rephasings. Its intrinsic triangle responses are*

$$T_{ijk} = \text{Tr}(P_i B P_j B P_k B) = B_{ij} B_{jk} B_{ki}.$$

They retain all loop-phase information. A simultaneous weak-basis change transports A, B , and all P_i and leaves each transported incidence response unchanged. A reversal of a triangle conjugates its response; weak CP conjugates all responses.

Proof. The trace formula follows by multiplication of the rank-one projections in any eigenframe of A . Cyclicity of trace proves its invariance under simultaneous unitary transport. Every loop response is a product of triangle responses, with inverse factors for reversed orientations. If two edge-phase representatives have identical triangle responses, their ratio has trivial triangle responses. Gauging the star at any temporary vertex to 1 makes every remaining ratio edge equal to 1; the ratio is therefore a vertex gauge. Conversely vertex gauges cancel around every loop. At a fixed simple A , its unitary stabilizer consists precisely of these diagonal rephasings. Thus no further phase information is identified. The temporary vertex is used only in the inverse proof. Hermiticity gives the reversal law, and complex conjugation gives the weak-CP law. These laws hold before the count is specialized. $\square$

Any admitted simultaneous law-only witness can serve in the neutral sector. The displayed choice establishes existence: it supplies nontrivial boundary transfer with a real neutral witness and no additional leptonic CP phase at that point. Different mechanisms remain possible wherever they satisfy the same declared primitive duties.

4.2 The explicit three-position joint point

Proposition 4.4 (One parameter point realizes all primitive duties). *The shared packet with*

$$A = \text{diag}(1, 4, 9), \quad B = \begin{pmatrix} 10 & 1 & 1 \\ 1 & 20 & i \\ 1 & -i & 30 \end{pmatrix}, \quad C = \text{diag}(1, 9, 25)$$

and the charged Yukawas above admits the simultaneous neutral choice

$$O = \frac{1}{7} \begin{pmatrix} -6 & 2 & 3 \\ 2 & -3 & 6 \\ 3 & 6 & 2 \end{pmatrix}, \quad m_\nu = \mu O \text{diag}(1, \sqrt{2}, 2) O^T, \quad \kappa = 2m_\nu/v^2.$$

It supplies a joint witness for representation response, charge compatibility, weak chirality, nondegenerate mass response, flavor participation, boundary reconstruction, and live quotient-level CP/holonomy. Its construction uses no empirical family selector, extra primitive fermion, or independent parameter point spliced from another witness.

Proof. The gauge and charged-sector claims are the preceding theorem's simultaneous equations. The intervals $[8, 12]$, $[18, 22]$, $[28, 32]$ make B positive and simple. Its three pair responses equal 1, its triangle response is i , and

$$\text{Tr}[A, B]^3 = 720i.$$

Thus the orientation response survives rephasing and basis transport. The matrix O satisfies $OO^T = I$ and $\det O = 1$. Its neutral response is

$$\mu^{-2} m_\nu^* m_\nu = \frac{1}{49} \begin{pmatrix} 80 & 48 & 30 \\ 48 & 166 & 18 \\ 30 & 18 & 97 \end{pmatrix} =: N_0.$$

The spectrum is $(1, 2, 4)$, and every off-diagonal entry is positive. For $U_0(t) = e^{-itN_0}$ and $0 < t \leq 1/100$, the integral remainder is at most $8t^2$; using the weaker $16t^2$ gives

$$|(U_0(t))_{ji}| \geq \frac{18}{49}t - 16t^2 \geq \frac{254}{1225}t > 0 \quad (i \neq j).$$

All channels use the same N_0 , source and detection projectors, v , μ , and propagation parameter. Charged and neutral mass terms use the same Higgs fixation. The Weinberg contraction is

gauge invariant and adds no primitive source node. The tensor equations therefore coexist at one parameter point; no incompatible parameter witnesses have been combined. All declared compositions are evaluated on this one packet, with undefined or failed uses retained as such. $\square$

4.3 Generated uses, failure records, and joint coherence

Proposition 4.5 (One packet supports the declared use closure). *Fix one of the constructed packets. Take as generators its representation and gauge maps, charged and neutral tensors with their actual source and target types, intrinsic occurrence projectors, direct and loop responses, and the declared source–propagation–readout maps with their parameter domains. Close this repertoire under finite typed composition, tensor contraction, adjoint, endomorphism trace, and inverse on the actual invertibility domain. Retain ill-typed or undefined uses as a distinguished failure value.*

Simultaneous lawful basis transport preserves the domains and interpreted values of all such generated uses, with the appropriate covariance for vector- or operator-valued outputs. Failure status is preserved. Evaluated histories can be retained by append-only records, without replacing a previous successful or failed event.

Proof. Every generator belongs to the same common packet, so its input and output spaces, parameters, and incidence are fixed jointly. Gauge and basis transport act on the representation, charged, and neutral tensors by their respective source, target, and dual actions. In particular, a Yukawa map need not transform by conjugation, and a symmetric Weinberg tensor has its own congruence action. The generator equations and their typed domains are covariant.

Composition cancels matching intermediate basis maps; contractions respect the dual pairing; adjoints use unitarity; and endomorphism traces are conjugation invariant. Invertibility is invariant under invertible changes of source and target basis, and the inverse has the reversed source–target action. Induction on a generated expression proves covariance and domain preservation. Assigning the failure value when a type or domain test fails therefore gives an invariant status.

A history is a finite ordered sequence of evaluated event records, including their interpreted incidence, domain status, and value or failure. The update appends the next event. An earlier record remains a prefix under this operation. Transport re-expresses the whole recorded incidence coherently and preserves its interpreted content; it does not overwrite a past event with a revised answer. $\square$

This proposition establishes the generated-use and failure clauses for the declared repertoire. Extending the repertoire to additional physical continuations requires their warrants and the same joint compatibility test.

4.4 Eligibility, homogeneous positions, and joint occupancy

The structural role construction supplies three homogeneous occurrence positions R . For the joint point above, their occurrence representatives are the intrinsic spectral line projections of A :

$$X = \{P : P = P^* = P^2, \text{ rank } P = 1, [P, A] = 0\}.$$

Because A is simple, X is exactly its three spectral projections. Each position is eligible for each member of X : the representation, charge and weak response types are homogeneous; mass response is nondegenerate; every pair comparison is nonzero; and the sole triangle has

live holonomy. Boundary and common-cone duties belong to the same shared packet, not to independently assigned channel parameters.

Joint compatibility retains that packet and requires pairwise orthogonality, resolution of identity, all actual transition and boundary maps, and the common parameter/charge/anomaly/failure records. Every bijection $\sigma : R \rightarrow X$ satisfies these conditions. There are no additional spectral-line candidates: a rank-one projection commuting with simple A is one of its spectral projections.

A permutation α of the homogeneous role names acts by $\sigma \mapsto \sigma\alpha^{-1}$ and transports role-indexed queries with it. It fixes the physical records and is transitive on all complete assignments. It does not assert a physical stabilizer exchanging unequal masses. An orientation coordinate is transported in its torsor; no external orientation pole is selected.

The common multiplicity is realized in every sector, with each sector's tensors carrying their own lawful basis actions and retaining their response data. The received Standard Model plus Weinberg inventory contains no pairing-sensitive map between quark and lepton mass eigenlines. Any displayed pairing of their labels is presentation data. Generation eligibility and joint compatibility therefore depend on the shared multiplicity and actual sector tensors, independently of numerical eigenvalue matching.

Positive certificate construction. With the primitive role object derived in Corollary 5.2 and Theorem 3.9, this packet supplies the positive generation certificate: all nine primitive duties have simultaneous witnesses, every eligible occurrence is covered, and the complete assignment is unique modulo lawful role-incidence transport. The structural role theorem establishes primitive minimality, and the displayed matrices establish joint realization. Together they give $\text{ClosedGenCert}_{\mathcal{P}_0}(3)$ as defined in Definition 5.3.

Joint realization on a minimal live comparison. Every live comparison class has a live triangle. Restrict its abstract phase class to that triangle and apply Theorem 4.2 at its derived arity. The resulting packet retains that triangular phase response and realizes all nonphase duties together. This construction establishes joint eligibility on each minimal live support. Its existential realization fields are supplied on that support, as in the constructional preassignment of Section 3.

5 The forced primitive role architecture and exact endpoint

The joint realization of Section 4 applies the minimality theorems to the complete primitive duty package in Definition 2.1.

5.1 Constructional candidates and compatibility

A constructional candidate is an anonymous occurrence set I with an admitted phase-comparison witness $q \in Q_I(\mathbb{R}/\mathbb{Z})$ satisfying $2q \neq 0$. The witness establishes nonvacuous reversal-sensitive capability. Ordinary non-removable phase data can also be two-torsion and hence reversal invariant. For the phase circle, the two capacity predicates of Definition 3.1 have the same minimal architectures, and the uniform receiving family supplies a reversal-sensitive witness wherever that capacity exists.

The resulting adequacy predicates have different quantifiers. *Capacity adequacy* of a bare architecture means that some admitted live comparison witness has a joint receipt. *Witness adequacy* retains one specified live class. CP-conserving parameter points remain possible

on the same capacity-bearing architecture. These statements compare admissible parameter assignments; a physical time evolution requires its own dynamical specification.

For a subset $J \subseteq I$, call $(J, q|_J)$ *witness-adequate* when $2q|_J \neq 0$ and there is one simultaneous realization of every duty in $\mathcal{P}_0$ on that support with exactly the retained comparison class. Receiving parameters are existential variables of the constructional problem, supplied jointly for each support. A fixed physical packet defines a separate preservation problem for its already specified records.

Theorem 5.1 (Uniform primitive adequacy). *Every finite reversal-sensitive phase-comparison class has a joint $\mathcal{P}_0$ realization. In particular, every live triangle in an arbitrary occurrence carrier has a joint primitive realization with its restricted comparison class. No support with fewer than three positions has that capability.*

Proof. The count-open construction in Section 4 realizes every finite phase class and checks representation, charge, weak, mass-slot, flavor, boundary, and whole-packet compatibility simultaneously. Its intrinsic loop responses recover the input comparison class, so a reversal-sensitive input supplies the required nonvacuous holonomy use. The construction uses no observed family labels, fitted textures, or preferred triangle.

Every live triangle is a finite instance of that theorem, so the entire primitive duty package can be received on it, not merely the Ω component. For fewer than three positions, Theorem 3.2 gives zero comparison quotient, excluding the required nonvacuous capability. $\square$

Corollary 5.2 (Minimum complete primitive construction). *The least occurrence cost of a complete $\mathcal{P}_0$ construction is three. Every support-minimal witness-adequate primitive comparison construction has three positions, and its bare incidence architecture is K_3 .*

Proof. Take any reversal-sensitive class. A live triangle exists, and Theorem 5.1 supplies all primitive duties on that support. A larger support is therefore not minimal for this constructional work. A three-position adequate support is minimal because no proper subset has nonvacuous comparison capacity. Nonemptiness follows, for example, by assigning a phase not equal to its inverse to one edge of a triangle. Theorem 3.5 then applies. This is a structural minimum argument; the numerical parameters used to receive it do not supply its lower bound or select its shape. $\square$

The least architecture is also the least capacity-bearing architecture, because the uniform receiving family introduces no further lower bound. This statement covers the possibility of larger or infinite candidate occurrence sets: a finite adequate minimum already exists. It does not postulate a finite upper bound on every enriched realization.

5.2 Roles formed before residue matching

Form the class of all support-minimal witness-adequate primitive completions, allowing their phase witnesses and receiving parameters to vary. Retain the bare homogeneous occurrence-position architecture. By Corollary 5.2, this class has one isomorphism type. Let $R\mathcal{P}_0$ be an abstract representative.

The target role set is formed before any candidate residue inventory is matched, and its count is calculated by the minimum theorem. A representative of the *whole architecture* retains its three positions and their incidence. The orbit set of individual vertices has one element, so it represents a different object.

For a live candidate (I, q) , a constructional preassignment is an injection

$$\sigma : R_{\mathcal{P}_0} \longrightarrow I$$

whose image is a minimal witness-adequate primitive support with the restricted phase data. The joint predicate means that the image admits one simultaneous $\mathcal{P}_0$ realization. All such preassignments are included. The live occupant set is the union of their images, as in role-occupancy closure [10].

Each preassignment supplies an alternative joint realization before fixation. Restrictions of a fixed enlarged physical packet have their own eligibility conditions. At a complete assignment the whole candidate is the support, and one receipt packet realizes every occurrence jointly.

Definition 5.3 (Closed primitive generation certificate). A closed primitive generation certificate at cardinality n consists of:

- (i) a constructional candidate (I, q) with $|I| = n$ and nonvacuous admitted holonomy capacity witnessed by q ;
- (ii) the target architecture $R_{\mathcal{P}_0}$ formed from the minimal complete primitive constructions, independently of that candidate;
- (iii) a compatible preassignment onto the entire live occupant union, with no omitted live occupant or added primitive role;
- (iv) one joint physical receipt for the complete assignment, retaining its actual incidence, response, boundary, and failure data, and satisfying the homogeneous lawful assignment-orbit condition.

Write $\text{ClosedGenCert}_{\mathcal{P}_0}(n)$ for existence of such a certificate.

Definition 5.3 counts complete occupation of the independently formed primitive role template. Its cardinality and circuit rank are derived below. An enriched receiver has a primitive count through its interpretation of this template and the corresponding occupation relation.

Theorem 5.4 (Generation cardinality as a forced primitive endpoint). *For the primitive generation construction specified by $\mathcal{P}_0$,*

$$\boxed{\text{ClosedGenCert}_{\mathcal{P}_0}(n) \iff n = 3.}$$

Its target role set has three positions. Its closed live residue set also has three positions. Complete homogeneous assignments form one lawful role-incidence orbit.

Proof. By Theorem 5.1, a minimal witness-adequate support is exactly a live triangle. Every occurrence of a live comparison class belongs to a live triangle: for a fixed live abc and an outside v , the tetrahedral identity

$$T(q)_{abc} = T(q)_{vbc} - T(q)_{vac} + T(q)_{vab}$$

forces at least one triangle on the right to be live. Thus the live occupant union of all constructional preassignments is I .

Corollary 5.2 gives $|R_{\mathcal{P}_0}| = 3$. A complete preassignment is an injection from that derived role architecture onto I , hence $|I| = 3$. This proves necessity without a proposed triangle selector.

Conversely, choose a live phase class on a three-position set. The entire set is a minimal witness-adequate support. The uniform joint receipt realizes all primitive duties on this same support; its intrinsic spectral inventory exhausts the physical occurrences. Every role bijection

is a compatible complete assignment. Precomposition by permutations of the homogeneous role set transports role-indexed questions with the assignment, fixes the physical packet and its interpreted records, and acts transitively. Thus the certificate closes. No transitive stabilizer of the numerically decorated physical packet is assumed. $\square$

If a role-specific orientation pole or other decoration is independently fixed, its actual preserving group replaces the bare permutation group. That decorated orbit claim needs its own check. For the displayed oriented triangle, the alternating subgroup acts transitively on matching orientations. The cardinality statement concerns the declared homogeneous primitive target.

5.3 Rank one is a consequence

Corollary 5.5 (Primitive circuit register). *The integral comparison quotient of the forced primitive architecture is isomorphic to $\mathbb{Z}$, and its phase quotient is isomorphic to $U(1)$. Its real comparison space has dimension one.*

Proof. On a triangle, quotienting the three oriented edge values by vertex gradients leaves their oriented sum. Fixing two tree-edge values to zero proves that this sum identifies the integral quotient with $\mathbb{Z}$, and the same argument over the phase group identifies its quotient with $U(1)$. Over $\mathbb{R}$, the quotient has dimension $3 - (3 - 1) = 1$. Reversing the orientation changes the sign or inverts the phase. A chosen orientation is a coordinate convention, not an additional primitive occupant. $\square$

For a finite complete graph on $n \geq 1$ positions, the corresponding free rank is $\binom{n}{2} - (n - 1) = (n - 1)(n - 2)/2$. Evaluating this expression on the architecture established by Theorem 5.4 gives rank one. The rank is a property of that architecture's comparison quotient; uniqueness of the bare architecture is the separate isomorphism statement in Corollary 5.2.

5.4 Primitive count and receiving inventories

Theorem 5.4 counts the positions of the minimal primitive architecture and its complete live occupation. The uniform construction supplies one simultaneous receipt of all duties, while the template and live-union arguments establish the cardinality of a complete assignment.

For an enriched realization, the primitive role inventory is obtained through the declared interpretation and occupancy relation. Its total physical inventory may also include positions required by further tasks. Section 7 gives the conditions under which a downstream domain receives the primitive endpoint.

6 Primitive standing and nonpromotion under enrichment

The preceding construction determines which architectures are necessary for the original generation work. Primitive standing can now be stated directly, before an occupation map is chosen. This separates the theorem that forces the role architecture from the theorem that counts its complete occupation.

6.1 A count-independent necessity test

Definition 6.1 (Primitive architectural standing). Fix the original primitive work $\mathcal{W}$, its homogeneous occurrence types, and its complete-comparison meaning. Write

$$A_{\mathcal{W}}(I) \iff \text{one joint construction of all } \mathcal{W} \text{ exists on } I.$$

The comparison witness and receiving parameters are existential construction data. Define

$$\text{Prim}_{\mathcal{W}}(I) \iff A_{\mathcal{W}}(I) \wedge (\forall J \subsetneq I, \neg A_{\mathcal{W}}(J)).$$

Adequacy on J retains every original duty of $\mathcal{W}$ and permits one joint construction of its receiving data there. The definition mentions no numerical count, circuit rank, selected triangle, or fitted physical value.

This is the minimal-necessity meaning of primitive construction. An occurrence architecture has primitive standing when its complete original work cannot be constructed on a proper support. A fixed physical packet can also be irreducible for its own dynamics; that is a different predicate because it holds the packet's additional data fixed. The quantified test examines every proper support. Single-vertex deletions alone need not establish minimality for an arbitrary joint predicate.

Theorem 6.2 (Exact primitive generation standing). *For the original generation work $\mathcal{P}_0$ and its received complete-comparison architecture, every occurrence set I satisfies*

$$\boxed{\text{Prim}_{\mathcal{P}_0}(I) \iff |I| = 3.}$$

Thus the possible primitive occurrence cardinality is exactly three: zero, one, two, every larger finite cardinality, and every infinite cardinality are excluded from primitive architectural standing. All primitive bare architectures have the isomorphism type K_3 .

Proof. Every adequate architecture must have the nonvacuous closed-comparison capability required by $\mathcal{P}_0$. On fewer than three positions every complete-graph cochain is a vertex gradient, so Theorem 3.2 excludes adequacy.

Proposition 4.4 supplies one simultaneous three-position construction of every duty. Lawful occurrence transport carries this receipt to every three-position architecture. All its proper supports have fewer than three positions and are inadequate. Hence every three-position architecture has primitive standing.

Conversely, an adequate architecture with more than three positions contains a proper triple J . The transported joint construction makes $A_{\mathcal{P}_0}(J)$ true, contradicting the necessity test in Definition 6.1. The same argument applies to an infinite I ; it uses the finite receipt on J and asserts no infinite-dimensional physical receiver. The complete simple graph on three positions has one bare isomorphism type. $\square$

The theorem classifies every architecture with primitive standing for the complete original task. The numerical data in the positive receipt establish simultaneous adequacy; they neither select the count nor enter the comparison obstruction that supplies the lower bound.

Corollary 6.3 (Specified witnesses and complete occupation). *For a live constructional candidate (I, q) with a joint $\mathcal{P}_0$ receipt, the following are equivalent:*

1. *no proper $J \subsetneq I$ admits the complete primitive work with the restricted comparison class $q|_J$;*

2. $|I| = 3$;

3. *one compatible primitive preassignment covers the entire live union of the constructional preassignments.*

Consequently the primitive-standing classification and the closed occupation theorem give the same exact count.

Proof. Triangle detection and uniform joint reception supply a proper adequate live triangle whenever I is larger than three. At three, every proper subset has zero comparison quotient. This proves $(1) \Leftrightarrow (2)$. Theorem 5.4 proves $(2) \Leftrightarrow (3)$, including live-union coverage and the positive one-packet assignment. Its role template is formed from the already classified minimal architectures, independently of the candidate inventory. $\square$

The three positions are homogeneous occurrence positions of one architecture. They are not three different unary work-types. Whole architectures are compared up to lawful incidence isomorphism; taking the orbit set of individual vertices would erase positions and count a different object. Replacing an occupant, choosing another compatible witness, or realizing several copies also differs from adding a position to one primitive architecture.

6.2 Additional downstream structure supplies no extra primitive role

Theorem 6.4 (Nonpromotion for unchanged primitive work). *Keep $\mathcal{P}_0$ and its original constructional meanings fixed. Let an enlarged realization propose an occurrence architecture I' as the complete primitive generation architecture. If $|I'| > 3$, then*

$$\neg \text{Prim}_{\mathcal{P}_0}(I').$$

Additional consistent values, responses, executions, or interactions cannot confer primitive standing on that enlarged architecture for the unchanged work. This conclusion also holds when the enlarged packet is irreducible under all its own physical operations.

Proof. If the proposed architecture fails a duty of $\mathcal{P}_0$, it lacks primitive standing by Definition 6.1. If it satisfies the duties, its proper three-position support admits a complete joint construction of that same work by Theorem 6.2. It therefore fails primitive necessity. Keeping additional enriched data fixed can change the enriched construction problem; it does not change $A_{\mathcal{P}_0}$ while the original work and meanings remain fixed. $\square$

The conclusion concerns additional *simultaneous primitive roles*. Every occurrence of a larger live carrier may participate in some alternative triangular witness. Such alternative eligibility does not make their union one minimal architecture or give one complete assignment covering that larger union. No particular ambient occurrence is singled out as the surplus one.

Additional responses can retain standing within the physical task that requires them. Nonpromotion denies their acquisition of independent primitive generation authority through downstream existence alone. If a proposed requirement was indispensable original primitive work, its receipt belongs in $\mathcal{P}_0$'s interpretation ledger. The theorem does not authorize dropping such a requirement by renaming it an enrichment.

Proposition 6.5 (Invariance under work-equivalent formulations). *Suppose two independently interpreted formulations $\mathcal{W}$ and $\mathcal{W}'$ have the same adequate architectures:*

$$A_{\mathcal{W}}(I) \iff A_{\mathcal{W}'}(I) \quad \text{for every occurrence architecture } I.$$

Then $\text{Prim}_{\mathcal{W}}(I)$ and $\text{Prim}_{\mathcal{W}'}(I)$ agree for every I . Their primitive cardinalities and bare minimal architectures agree.

Proof. Substitute the adequacy equivalence for I and each of its proper supports in Definition 6.1. $\square$

This is the precise invariance under a faithful reformulation of the primitive work. A new requirement that changes the adequate-architecture family must be assessed as a changed question or an identified omission from the claimed original specification.

6.3 Count descent without a preferred embedded witness

Let $\mathfrak{P}_0$ be the primitive constructions classified by Theorem 6.2, retaining their original interpretations and occupation conditions. An enriched object e can receive such a construction by a function or by a licensed relation

$$Z \subseteq \mathcal{E} \times \mathfrak{P}_0.$$

Write $Z(e) = \{p : (e, p) \in Z\}$ and $\mathcal{E}_Z = \{e : Z(e) \neq \emptyset\}$. The relation records the actual claimed primitive receipts. Containment of an abstract triangle alone does not supply a faithful interpretation of every use of a fixed physical parent.

Theorem 6.6 (Relational descent of primitive cardinality). *Every $e \in \mathcal{E}_Z$ has the unique primitive count*

$$N_{\text{prim}}(e) = 3,$$

independent of the receipt $p \in Z(e)$. No function choosing a primitive representative, embedded triangular subcarrier, or orientation is required. If physical occupants are claimed, a jointly eligible injective assignment from the received primitive architecture that covers the claimed primitive occupant set makes that set have exactly three members.

Proof. Every $p \in Z(e)$ has three positions by Theorem 6.2. Thus every possible receipt gives the same count; nonemptiness makes the count an actual receipt value rather than a vacuous condition. An injective complete occupation is a bijection from this three-position architecture to the claimed primitive occupant set. $\square$

The selector obstruction in Proposition 3.10 is therefore compatible with a unique primitive count. A four-position symmetric carrier need not distinguish one of its triangular witnesses. Its full witness relation still has constant primitive arity. A function selecting a physical triangle is a further assertion with a different burden.

Proposition 6.7 (Exact reader descent along a receipt relation). *Let $k : \mathfrak{P}_0 \rightarrow Y$ be an independently specified original reader, with any required admission and failure status included in its outcome. There exists $h : \mathcal{E}_Z \rightarrow Y$ satisfying*

$$h(e) = k(p) \quad \text{whenever } (e, p) \in Z$$

if and only if k is constant on every nonempty fibre $Z(e)$. When it exists, h is unique. If an enriched reader h_0 is already prescribed, conservation requires this equality with that h_0 .

Proof. An h with the displayed property equates the values of k at any two receipts of e . Conversely, fibre constancy and nonemptiness give a unique value of k for each e ; assign that value to $h(e)$. Any compatible h must take that value. An independently prescribed h_0 has its fixed meaning and must equal it. $\square$

When every original reader is constant on each receipt fibre, the proposition gives a unique map into the quotient of primitive constructions by complete original-use profile. It need not select a representative construction. Primitive count always passes this test on $\mathfrak{P}_0$; other answers need not. For example, two compatible constructional witnesses can have different reversal-orbits of triangle phase while both have three positions. Their relation determines one count but does not determine that phase answer. A witness relation for capability and a complete representation of one fixed phase-bearing object make different claims.

6.4 Grounded construction cannot add independent primitive authority

The count theorem identifies primitive necessity. A receiving derivation must also conserve the original grounds of qualification [4, 7, 9]. Fix a set F of statements and a finitary positive rule system $L_0 \subseteq \mathcal{P}_{\text{fin}}(F) \times F$, with original grounds $S \subseteq F$. A rule (A, b) concludes b when all members of the finite premise set A have been established. Write $C_L(S)$ for the conclusions of finite proof trees using rules L and leaves in S . For each added finite macro rule (A, b) , suppose an explicit finite L_0 derivation establishes b from A .

Theorem 6.8 (Conservative derived authority). *Adding any such macros does not change the original finite derivability closure:*

$$C_{L_0 \cup M}(S) = C_{L_0}(S).$$

In particular, an enriched derivation cannot obtain a new original primitive-generation qualification solely from those macros. If auxiliary data are supplied as new grounds, the same conclusion holds when their asserted original statements already belong to $C_{L_0}(S)$.

Proof. Every L_0 derivation remains available after the extension, giving one inclusion. For the other, induct on a finite derivation in the extended rules. An original-rule step uses the induction hypotheses for its premises. At a macro step, replace that step by the supplied finite L_0 derivation and substitute the already obtained derivations of its premises. Thus its conclusion belongs to $C_{L_0}(S)$. Grounds already in the old closure can be replaced by their old derivations in the same way. Qualification conclusions obey the same argument as every other original statement. $\square$

The argument allows new descriptions, useful derived operators, and longer executions. It forbids obtaining new independent primitive authority from a derivation whose entire original content expands into the prior grounds. Introducing a new independent premise requires its own warrant; merely placing it at a later construction stage does not prove conservativity.

6.5 Capacity through physical fixation

The architectural theorem is stated before a particular realized phase is selected. A capacity receipt comprises an admitted realization family and a live witnessing point q_* in that family. A current physical point q_{cur} may instead satisfy $2q_{\text{cur}} = 0$. The admitted family still establishes the architecture's capacity and primitive arity. The standing of a particular reversal-sensitive use is evaluated at its actual point.

For the phase-circle receiver, the uniform construction receives both q_* and q_{cur} with their proper outcomes. They are alternative permitted parameter points, not two simultaneously imposed values of one event. No physical time evolution between them is claimed without a separate dynamical law. Thus a CP-conserving point does not invalidate the prior exact-three architectural theorem, and a capability witness is not substituted for a measured result.

7 Receiving the primitive endpoint

7.1 Primitive interpretation and physical preservation

An enriched realization may contain several overlapping live triangles, shared occurrence identities, and additional responses required by its physical task. Theorem 6.2 determines every architecture with primitive standing for $\mathcal{P}_0$. Theorem 6.4 excludes all larger architectures from that standing. The receiving task retains its own complete response data; their additional existence supplies no extra simultaneous primitive role.

For an orthogonal projection P and a self-adjoint response H , the identity

$$PH^2P - (PHP)^2 = PH(1 - P)HP$$

exhibits the contribution of propagation through the complementary subspace. Preserving this two-step response is an obligation of a task that includes it. In the primitive minimum problem, all retained duties are those fixed in $\mathcal{P}_0$. Constructional adequacy and preservation of a fixed physical packet can thus be assessed using their respective declared answers.

A primitive interpretation specifies its admission domains and use readers. Proposition 3.11 characterizes functional conservation by factorization. For relational receipts, Theorem 6.6 establishes the unique primitive count and Proposition 6.7 gives the exact condition for each additional original answer. Proposition 3.10 explains why a canonical function selecting a physical triangle need not exist. No such selector is required for the primitive count.

7.2 Criteria for applying the theorem

Application of the endpoint requires five checks.

1. *Primitive-duty coverage.* Identify the independently specified primitive work and establish that $\mathcal{P}_0$ receives all of it. Any additional original primitive duty must satisfy the same joint-reception requirement.
2. *Comparison structure.* Establish complete direct-comparison incidence and the quotient by vertex rephasing, with a physical response for every declared edge and phase. Sparse comparison structures require their own minimality classification.
3. *Joint realization.* Verify the representation, anomaly, mass-slot, transition, boundary, and assignment equations on one common packet.
4. *Minimality and occupation.* Verify exclusion below three, joint adequacy of the minimal live supports, and complete occupation of the full live union of the admitted capacity witness by the independently formed role template. A CP-conserving current point is evaluated separately from that witnessing point.
5. *Receiving interpretation.* Specify the counted inventory and its interpretation of the primitive roles, retaining the actual domains, eligibility conditions, and lawful assignment action.

These checks state the hypotheses on which the endpoint depends. A missing primitive duty, an inequivalent comparison quotient, a failed joint equation, or an adequate counterexample to the minimum or occupancy statement identifies the corresponding point of failure. Additional enriched outputs are assessed within the task that specifies them.

7.3 Theorem interfaces

Payload	Result available	Receiving obligation
Exact primitive standing	Theorem 6.2: primitive architecture iff three positions	Retain the full original duty package and complete direct-comparison meaning
Closed role endpoint	Theorem 5.4 and Corollary 6.3: exact-three complete occupation	Identify the primitive roles and cover the complete live occupant union
Nonpromotion	Theorem 6.4: larger architectures have no primitive standing for the same work	Keep the original task fixed; do not omit an indispensable original duty
Primitive rank	Corollary 5.5: one circuit generator	Use the integral, real, or phase meaning on the derived architecture
Joint realization	Theorem 4.2 and Proposition 4.5	Receive the licensed types and declared boundary model; retain actual domains and joint equations
Relational primitive count	Theorem 6.6: count three without a preferred witness	Supply a nonempty licensed receipt relation and the claimed occupation map
Original reader conservation	Proposition 3.11 and Proposition 6.7	Prove functional factorization or constancy on the actual receipt fibres
Grounded derived authority	Theorem 6.8: original derivability closure is unchanged	Expand each added finite macro into original rules; justify any additional grounds

Identifying an ambient inventory with the primitive generation positions requires an occupation theorem for that inventory. Physical parameter values, ultraviolet singlet counts, spectral and terminal inventories, and additional physical laws retain their separate domain conditions.

7.4 Applications to generation-governed constructions

The FCGS-I, SM-6, and BCRS-I constructions use a generation-governance input. Theorem 5.4 supplies that input when their counted objects receive the primitive duty package and its complete occupation relation. A proposed fourth simultaneous primitive occupant fails the exact primitive-standing and complete-occupation theorems for this original work. Merely enlarging a downstream matrix, response algebra, or physical inventory does not alter that result. The receiving proof establishes which actual objects occupy the primitive positions; it does not infer this identification from their shared numerical count.

FCGS-II/III and the CSP-P/R/S constructions receive the endpoint through their input structures, with source, response, terminal-bearer, marking, and transport data retained. PCR-5 separately specifies ultraviolet singlet occupancy; SM-5 states its flavor results on a declared generation envelope. These distinctions identify the counted object in each application.

A formal representation of the theorem consists of the primitive duty and assignment structures, the minimum construction, and the complete-occupancy proof. Their definitions precede the cardinality conclusion. The accompanying dependency map records the source-specific interpretation requirements and locations for these applications.

7.5 Conclusion

The complete original generation work determines its primitive occurrence architecture. The comparison obstruction excludes fewer than three positions; a common receiving packet proves joint adequacy at three; and every larger adequate architecture contains a proper adequate primitive construction. Hence

$$\text{Prim}_{\mathcal{P}_0}(I) \iff |I| = 3.$$

Roles are formed from this derived architecture before candidate matching. Their complete live occupation consequently satisfies

$$\text{ClosedGenCert}_{\mathcal{P}_0}(n) \iff n = 3.$$

Its primitive circuit register has rank one.

Downstream enlargement does not supply additional primitive generation standing for the unchanged work. A nonempty relation of licensed primitive receipts gives the same count without a preferred embedded witness. Preserving further original answers requires their receipt-fibre agreement, and derived authority remains grounded in the original qualification closure. Together these results establish primitive necessity, exact occupation, and nonpromotion in the order needed by receiving constructions.

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